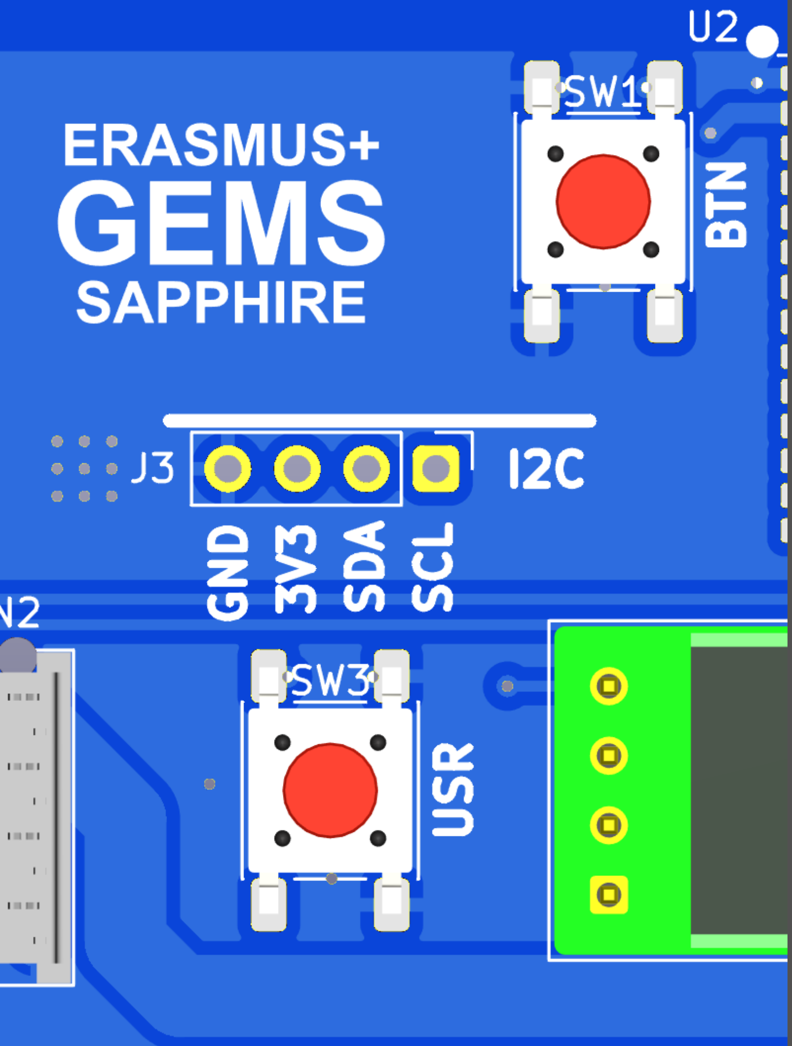

Sustainable Management of Mechatronics Hardware

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Objectives

1. Understand the Environmental Impact of Mechatronic Systems
2. Introduce Circular Economy Principles in Engineering Design
3. Identify Sustainable Design Strategies for Electronics and Prototyping
4. Promote Responsible Use and End-of-Life Handling of Educational Hardware

Why Sustainability Matters in Mechatronics

- Mechatronics = electronics + mechanics + software → real environmental impact
- Electronics waste is one of the fastest-growing waste streams
- Engineers influence impact at the design stage

What Is Circularity?

- Linear vs Circular Model
  - Linear: Take → Make → Use → Dispose
  - Circular: Design → Use → Reuse → Repair → Recycle
- Engineering perspective:
  - Keep materials and components in use longer
  - Reduce waste and energy consumption

Environmental Impact of Mechatronic Systems

- Where impact comes from:
  - PCB manufacturing (chemicals, copper, energy)
  - Electronic components (rare materials)
  - Motors, batteries, power electronics
  - Plastics (3D printing, enclosures)
- Key takeaway:
  - Most environmental impact is locked in during design, not during use.

Sustainable Design Strategies (Big Picture)

Intro-level design principles:

- Design for long life
- Design for repair
- Design for disassembly
- Avoid unnecessary complexity
- Prefer standard components

Electronics & PCB Design for Sustainability

- **Good practices for students:**
  - Minimize PCB size and layer count
  - Avoid exotic materials when possible
  - Use through-hole parts for learning boards (repairable)
  - Label boards clearly (version, function)
- **Educational mindset:**
  - Prototypes ≠ disposable trashTreat PCBs as reusable assets

Handling Electronic Components in Education

Before disposal:

- Desolder and recover reusable components
- Sort components (ICs, passives, connectors)

Storage & reuse:

- Organize salvaged parts
- Reuse in labs, demos, or future projects

Important lesson for students:

- Components have value even after a project ends.

3D Printing & Mechanical Parts

- Environmental considerations:
  - Material waste from failed prints
  - Energy usage of printers
- Better practices:
  - Optimize designs before printing
  - Print only what is neededPrefer PLA for educational use
  - Reuse printed parts in future projects
- Design tip:
  - Modular mechanical designs → parts can be reused

Batteries, Power & Energy Efficiency

- Common student mistakes:
  - Oversized power supplies
  - Leaving systems always ON
  - Poor power management
- Sustainable strategies:
  - Choose efficient motors and drivers
  - Use sleep modes and power control
  - Avoid disposable batteries when possible

What to Do with Old Educational Hardware

Do not treat as waste immediately:

Use for:

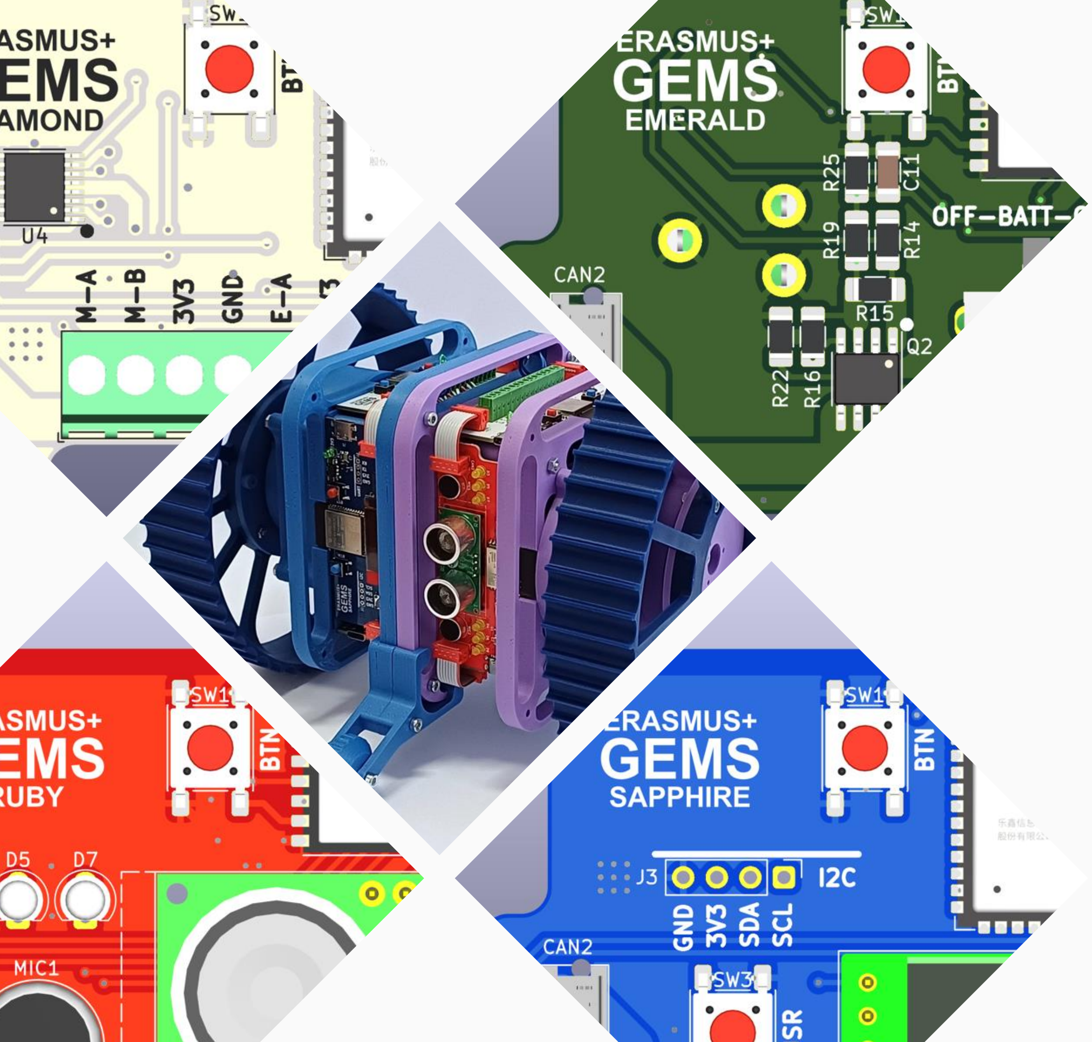
- Disassembly exercises
- Debugging practice
- Component harvesting
- Demonstration units

End-of-life thinking:

- Separate electronics, plastics, metals
- Follow proper e-waste channels

Engineer’s Responsibility

- Sustainability is not an add-on
- Every design decision has consequences
- Small design choices scale to big impact



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Thank you for watching!

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